

lations, and has set up a website to enable astronomers to track the satellites.

Space debris

With the continuing deployment of thousands of satellites into space, the space junk is going to multiply, increasing the possibility of catastrophic satellite collisions. Debris in space can be a threat to satellites and space stations in orbit, as well as future manned missions to space.

Also, the orbiting satellites eventually get pulled out of the orbit by the natural causes, such as the atmospheric drag experienced. The international standards require satellites to re-enter Earth's atmosphere within 25 years of mission completion, if they are in the LEO region. However, scientists and the experts want the 25-year time to be shortened.

Also, the question remains as to what happens if a satellite dies early. Starlink is considering de-orbiting satellites nearing the end of their useful lives by propelling them to a 'disposal orbit' for orderly re-entering the Earth's atmosphere within one year.

OneWeb maintains that their satellite networks are designed to avoid collisions with space stations, and they actively and regularly screen for conjunctions between their satellites and other space objects, catalogued by the Joint



Elon Musk-led SpaceX planning 42,000 satellites in orbit will play a key role in mitigating the hurdles ahead (Source: <https://www.republicworld.com>)

Space Operations Center (JSpOC). However, to obviate the possibility of a satellite hitting something else to create debris, they maintain that the probability of such collisions is compliant with NASA's Technical Standard (probability to be < 0.01).

Notwithstanding all such assurances, the European Space Agency (ESA), for the first time ever, had to perform a "collision avoidance manoeuvre" in September 2022 to protect one of its live satellites from colliding with a "mega constellation." 49 Starlink satellites launched from NASA's Kennedy Space Centre, Florida, suffered an early death due to a geomagnetic storm, pushing up the density of the atmosphere increasing the drag on the satellites.

Endpoint

Space based internet has been in use for several years using satellites in geostationary orbit. However, the use has been limited due to the huge cost involved and high latency.

The present-day satellite internet using satellites in LEO region involves lower latency and costs, allowing wider applications and usage. A

constellation of many satellites is required to provide continuous global coverage.

Starlink operated by SpaceX has become a space "juggernaut." As of now, it has 3,858 satellites in low Earth orbit, with a possible future extension to 42,000 satellites, providing coverage to

more than one million subscribers in over 50 countries. Starlink controls manufacturing, rocket launching, ground station construction and satellite operations.

In India, several players are active in providing seamless connectivity, and the satellite internet segment has become a new battleground for satellite internet operators and communications service providers. The government is yet to come out with the spacecom policy of India to lay down the rules and spectrum bands.

There are many challenges facing the satellite internet service providers, such as constraints in making available thousands of satellites simultaneously; maintaining the satellites in correct position in space; network problems caused due to bad weather conditions, such as heavy rainfall, thunder, and storms resulting in slower speeds or outages; and solving the problem of increased space debris along with addressing other concerns of astronomers.

Despite the challenges, satellite internet is moving ahead at full force, with rapid new advancements and applications. As of now, the satellite based internet services are complimenting the existing cable based infrastructure. It will be interesting to see if in the future this technology can improve and bring down the several limitations. **EFY**



The Joint Space Operations Center (JSpOC) is primarily a space traffic management centre (Source: <https://www.thespacereview.com>)